MoS2 layer thickness in SiO2-MoS2 TENGs

"How does MoS2 layer thickness affect dielectric loss in SiO2-MoS2 TENGs?" *Generated by LeapSpace*

MoS_2 layer thickness likely affects dielectric loss in SiO_2–MoS_2 TENGs nonlinearly: a thin-to-moderate MoS_2 layer should help suppress loss by improving charge trapping and blocking charge drift, but an overly thick layer can raise loss again by creating a more conductive, aggregated MoS_2 network. [11] [34] [104] [105]

What the evidence supports

MoS_2 thickness change	Expected effect on dielectric loss	Why
Thin to moderate increase	**Lower dielectric loss / better charge retention**	MoS_2 is repeatedly described as a charge-trapping layer that helps prevent charge dissipation, and the MoS_2/SiO_2 interface supports charge trapping and retention. [11] [34] [104]
Excessive thickness	**Higher dielectric loss risk**	Too much MoS_2 can form a conductive network or aggregated layers, which increases charge leakage pathways and weakens the insulating behavior needed for low loss. [104] [105]
Very thin layer	**May be insufficient to suppress loss fully**	If the MoS_2 layer is too thin, the charge-trapping and shielding effect is weaker, so charge drift/dissipation is less effectively blocked. [11] [104]

How this maps to SiO_2–MoS_2 TENGs

- The **SiO_2 shell** is the main insulating component, while MoS_2 mainly adds **charge trapping and interfacial polarization**. [11] [34] [106]
- Increasing MoS_2 thickness likely strengthens the MoS_2 contribution at first, which can improve charge capture and reduce dielectric loss-related dissipation. [34] [104]
- Past an optimum, however, thicker MoS_2 can shift the balance toward **higher conductivity and charge transport**, which is unfavorable for dielectric loss. [104] [105]
- The best-supported design target is therefore **a thin, continuous, well-dispersed MoS_2 layer on SiO_2**, not a thick MoS_2-rich shell. [11] [34] [104]

Bottom line

For **SiO_2–MoS_2 TENGs**, MoS_2 thickness is best treated as an **optimum-search variable**: increasing thickness should help **up to the point where charge trapping and interfacial polarization are maximized**, but beyond that, dielectric loss is likely to worsen because of **aggregation and conductive-network formation**. That conclusion is an inference from related MoS_2/TENG and core-shell dielectric studies, not a direct thickness sweep on one exact SiO_2–MoS_2 TENG system. [11] [34] [104] [105]

Confidence level: **Medium** — the direction of the trade-off is well supported by related core-shell and MoS_2-TENG evidence, but the retrieved sources do not include a direct MoS_2-thickness series for one exact SiO_2–MoS_2 TENG formulation.

References

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